

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester IV Examination, 2025-26 (NEP)

COMPUTER SCIENCE**Paper Code: CSCMJ45 / CSCMN41 : Numerical Computing**

Time: 2:30 Hours

Maximum Marks: 70

*Instructions: Attempt any 5 questions.***Question 1.** Derive the general term for evaluation and hence write the algorithm for: [14]

- (a) Polynomial Evaluation
- (b) Polynomial Division

Question 2. Find a real root of the equation $x^3 + x^2 + x + 7 = 0$, correct to 4 decimal places using: [7 + 7]

- (a) Regula-Falsi method
- (b) Newton-Raphson method

Compare the number of steps taken in each case.

Question 3. Solve the following system of linear equations by (a) Gauss-elimination and (b) Gauss-Jordan methods: [7 + 7]

$$\begin{aligned}9x_1 + 2x_2 + 4x_3 &= 20 \\x_1 + 10x_2 + 4x_3 &= 6 \\2x_1 - 4x_2 + 10x_3 &= -15\end{aligned}$$

Question 4. Decompose the matrix

$$A = \begin{bmatrix} 5 & 2 & -1 \\ 7 & 1 & -5 \\ 3 & 7 & 4 \end{bmatrix}$$

into the form LU and hence solve the system $AX = b$ where $b = [4 \ 8 \ 10]^T$. [7 + 7]

Question 5. From the table given below, calculate $f(1.023)$ and $f(1.235)$ using Newton-Gregory forward and Lagrange interpolation formula respectively: [7 + 7]

oprule x	1.00	1.05	1.10	1.15	1.20	1.25
$f(x)$	0.682689	0.706282	0.728668	0.749586	0.769861	0.788700
ottomrule						

Question 6. Solve the given differential equation $\frac{dy}{dx} = x^2 + y$ with $y(0) = 1$, to compute $y(0.25)$ using step size $h = 0.05$: [7 + 7]

(a) Euler's method

(b) Heun's method

Compare the results obtained in each case.

Question 7. Answer the following: [7 + 7]

(a) Calculate the following integral using Simpson's 1/3 rule with $h = 0.25$:

$$\int_0^2 \frac{1}{x^3 + x + 1} dx$$

(b) Integrate $f(x) = 2x^3 - 3x^2 + 4x - 5$ from -2 to 4 using Gauss-Legendre 2-point formula.

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